INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME	
CLASS	
INDEX NUMBER	

CHEMISTRY **9647/01**

Paper 1 Multiple Choice **21 September 2015**

1 hour

Additional Materials: Data Booklet
Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **16** printed pages and **0** blank page.

Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

1 How many neutrons are present in 0.13 g of ^{13}C?
[L = the Avogadro constant]

A 0.06 L **B** 0.07 L **C** 0.13 L **D** 0.91 L

Amt of ^{13}C = $\frac{0.13}{13}$ = 0.10 mol
From Data Booklet, C has 6 protons, so ^{13}C has 13 − 6 = 7 neutrons

No of neutrons in 0.10 mol of ^{13}C = 0.10 × 7 × L = 0.07L Ans: B

2 Equimolar amounts of F_2 and BrO_x^- ions react with OH^- ions to produce three products: H_2O, F^- ions and BrO_4^- ions.

What is the value of x?

A 1 **B** 2 **C** 3 **D** 5

Reduction half eqn: $F_2 + 2e^- \rightarrow 2F^-$
1 mol of BrO_x^- gains 2 mol of electron to produce BrO_4^-.
Final oxid state of Br in BrO_4^- = +7, thus oxid state of Br in BrO_x^- = +5
Therefore, x = 3
Overall eqn: $F_2 + BrO_3^- + 2OH^- \rightarrow 2F^- + H_2O + BrO_4^-$
Ans: C

3 Which gas can be most easily liquefied by cooling and applying pressure?

A He **B** CH_4 **C** F_2 **D** NH_3

NH_3 has the most significant and strongest intermolecular interactions (hydrogen bonding) while the rest (He, CH_4, F_2) have weaker instantaneous dipole-induced dipole (id-id) interactions. Intermolecular attractive forces are formed between particles when gas is converted into liquid phase. Since NH_3 forms the strongest intermolecular attractions, therefore it is the easiest to liquefy.
Ans: D

4 Fresh fruits and vegetables are subjected to radiation from the radioactive decomposition of ${}^{60}_{27}\mathrm{Co}$ to eliminate harmful microorganisms to prolong their shelf-life. During the decomposition, a neutron in the nucleus decomposes to form a proton, which is retained in the nucleus and an electron which is expelled from the atom. No other particle is produced.

$${}^{60}_{27}\mathrm{Co} \rightarrow \mathbf{X} + e^{-}$$

Which row in the table correctly describes the nuclear make-up of ${}^{60}_{27}\mathrm{Co}$ and **X**?

	${}^{60}_{27}\mathrm{Co}$	**X**	
	Number of neutrons	Number of protons	Number of neutrons
A	33	26	31
B	33	28	32
C	60	26	34
D	60	28	32

No of neutrons in ${}^{60}_{27}\mathrm{Co}$ = 60 – 27 = 33
No of neutrons in X = 33 – 1 = 32
No of protons in X = 27 + 1 = 28
Ans: B

5 In which pair of molecules are the values of the bond angles the closest?

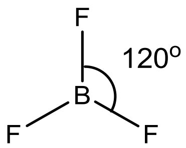
A BF_3 and NH_3

B C_2H_4 and BF_3

C H_2O and C_2H_4

D CH_4 and H_2O

A

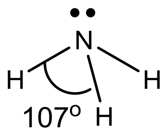


BF_3

3 bonding pairs & 0 lone pairs ard B

Shape: Trigonal Planar wrt B atom

Bond angle: 120°



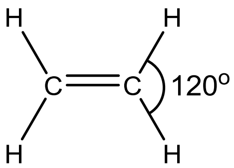
NH_3

3 bonding pairs & 1 lone pair ard N

Shape: Trigonal Pyramidal wrt N atom

Bond angle: 107°

B

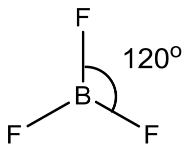


C_2H_4

3 bonding pairs & 0 lone pairs ard C

Shape: Trigonal Planar wrt C atom

Bond angle: 120°



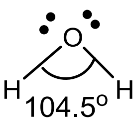
BF_3

3 bonding pairs & 0 lone pairs ard B

Shape: Trigonal Planar wrt B atom

Bond angle: 120°

C

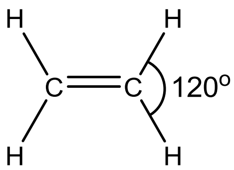


H_2O

2 bonding pairs & 2 lone pairs ard O

Shape: Bent/V-shape wrt O atom

Bond angle: 104.5°



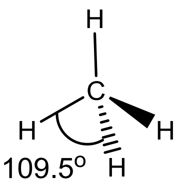
C_2H_4

3 bonding pairs & 0 lone pairs ard C

Shape: Trigonal Planar wrt C atom

Bond angle: 120°

D

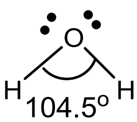


CH_4

4 bonding pairs & 0 lone pairs ard C

Shape: Tetrahedral wrt C atom

Bond angle: 109.5°



H_2O

2 bonding pairs & 2 lone pairs ard O

Shape: Bent/V-shape wrt O atom

Bond angle: 104.5°

Ans: D

6 In which change would only van der Waals' forces have to be overcome?

A evaporation of ethanol

B melting of ice

C melting of solid carbon dioxide

D solidification of butane

A evaporation of ethanol: **Hydrogen bonding** between ethanol molecules is broken

B melting of ice: **Hydrogen bonding** between H_2O molecules in ice is broken.

C melting of solid carbon dioxide: ***Van der Waals' forces of attraction*** between non-polar CO_2 molecules is broken.

D solidification of butane: ***Van der Waals' forces of attraction*** between non-polar butane molecules is broken.

Ans: C

7 Hess's Law can be used to calculate the average C–H bond energy in methane.

$\Delta H_{at}^{\ominus}$ = standard enthalpy change of atomisation

$\Delta H_{f}^{\ominus}$ = standard enthalpy change of formation

$\Delta H_{c}^{\ominus}$ = standard enthalpy change of combustion

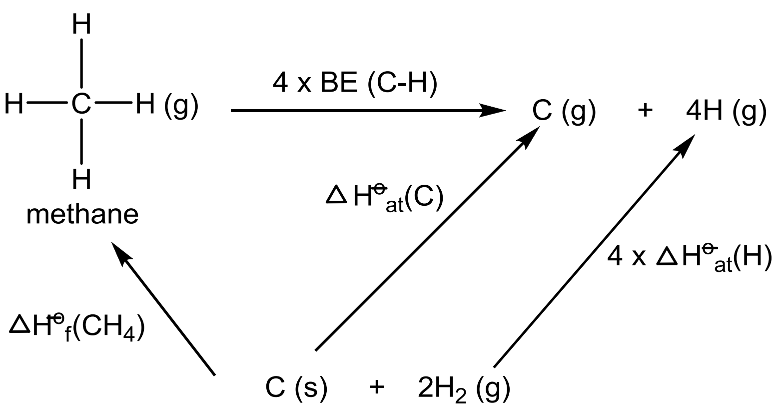
Which of the following information is needed in order to perform the calculation?

A $\Delta H_f^{\ominus}$ (CH_4) only

B $\Delta H_{at}^{\ominus}$ (C), $\Delta H_{at}^{\ominus}$ (H), $\Delta H_f^{\ominus}$ (CH_4)

C $\Delta H_c^{\ominus}$ (C), $\Delta H_c^{\ominus}$ (H_2), $\Delta H_c^{\ominus}$ (CH_4)

D $\Delta H_c^{\ominus}$ (C), $\Delta H_c^{\ominus}$ (H_2), $\Delta H_f^{\ominus}$ (CH_4)



Ans: B

8 Ethane undergoes combustion as follows:

$$C_2H_6(g) + 3\frac{1}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l) \qquad \Delta H_c^{\ominus} = -1561 \text{ kJ mol}^{-1}$$

Given the following data, calculate the standard enthalpy change of vapourisation of H_2O(l).

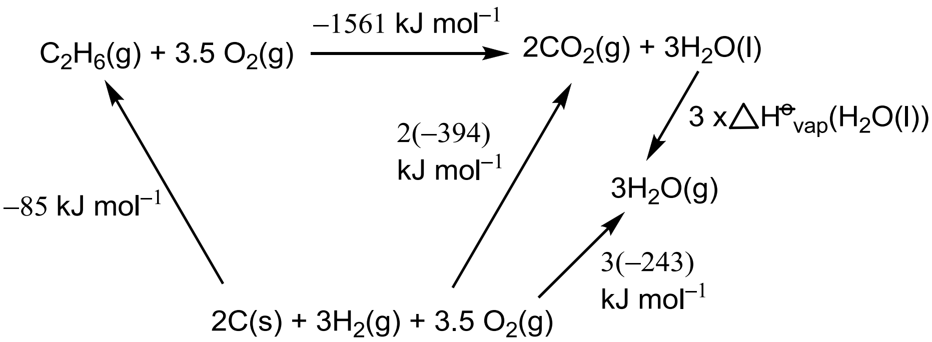
$\Delta H_c^\ominus$ of C(s) = –394 kJ mol^{-1}

$\Delta H_f^\ominus$ of C_2H_6(g) = –85 kJ mol^{-1}

$\Delta H_f^\ominus$ of H_2O(g) = –243 kJ mol^{-1}

A +43 kJ mol^{-1} **C** +129 kJ mol^{-1}

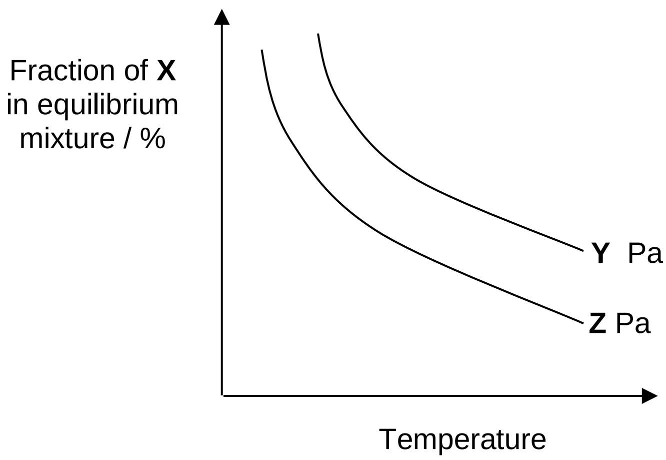
B –43 kJ mol^{-1} **D** –129 kJ mol^{-1}



By Hess Law,
–1561–85 = 2(–394)+3(–243) –3 ×$\Delta H_{vap}^\ominus$ of H_2O(l)
$\Delta H_{vap}^\ominus$ of H_2O(l) =**+43 kJ mol^{-1}**
Ans: D

9 The graph below shows how the fraction of a substance, **X** represented by one of the following compounds in the equilibrium mixture shown below varies with temperature at pressures of **Y** Pa and **Z** Pa.

$$4NH_3(g) + 3O_2(g) \rightleftharpoons 2N_2(g) + 6H_2O(g) \qquad \Delta H = -1267 \text{ kJ mol}^{-1}$$



Identify **X** and the correct relative magnitudes of **Y** and **Z**.

	X	Pressure
A	N_2	**Z** > **Y**
B	O_2	**Y** > **Z**
C	H_2O	**Y** > **Z**

D	NH_3	**Z** > **Y**

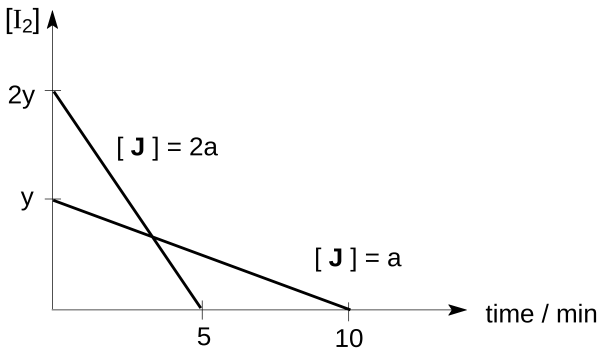
Answer:

From the graph, when the temperature increases for a particular pressure, the % of X decreases. As the equilibrium will shift to the left at higher temperature to favour the endothermic reaction X should be the product in the equilibrium i.e. either N_2 or H_2O.

Also, as the equilibrium will shift left when the pressure increases at a particular temperature, this means that a smaller % of X, the product, will be seen for a higher value of P. This means that the value of **Z** is greater than **Y**.

The answer is therefore **A.**

10 The kinetics of the reaction between iodine and compound **J** is investigated.



What conclusions can be drawn from the graphs?

A The reaction is second order with respect to compound **J** because rate of reaction increases by four times when its concentration is increased by two times.

B Both iodine and compound **J** are involved in the rate determining step.

C The reaction is first order with respect to iodine because half-life is constant.

D The overall order of the reaction is 1.

Answer:

When the [J] drops by half, the gradient of the graph drops by (2y/5) / (y/10) = 4 times. Hence the reaction is second order with respect to compound **J**. The answer is **A.**

11 A radioactive element has 2 isotopes, **G** and **H**, with half-lives of 2 days and 4 days respectively. An experiment starts with 4 times as many atoms of **G** as of **H**.

Radioactive decay is a first-order reaction.

How long will it be before the number of atoms of **G** left equals the number of atoms of **H** left?

A	8 days	**C**	16 days
B	32 days	**D**	64 days

Answer:

Let the number of atoms of **G** be 4x and the number of atoms of **H** be x.

Radioactive decay for **G** after 8 days (4 half-lives of 2 days):

4x → 2x → x → x/2 → x/4

Radioactive decay for **H** after 8 days (2 half-lives of 4 days):

x → x/2 → x/4

G goes through 4 half lives (8 days) before it can have the same amount of atoms as **H** at the same time. The answer is **A**.

12 Which species is dominant when equal volumes of aqueous solutions of 0.1 mol dm^{-3} H_2SO_4, 0.1 mol dm^{-3} NaOH and 0.1 mol dm^{-3} $H_2NCH(CH_3)COOH$ are mixed?

A	$H_2NCH(CH_3)COOH$	**C**	$H_3N^+CH(CH_3)COO^–$
B	$H_2NCH(CH_3)COO^–$	**D**	$H_3N^+CH(CH_3)COOH$

Answer:

Let the volumes of each solution be y dm^{-3}.

Moles of H^+ from H_2SO_4 = 2 x y x 0.1 = 0.2y mol

Moles of OH^- from NaOH = 0.1y mol

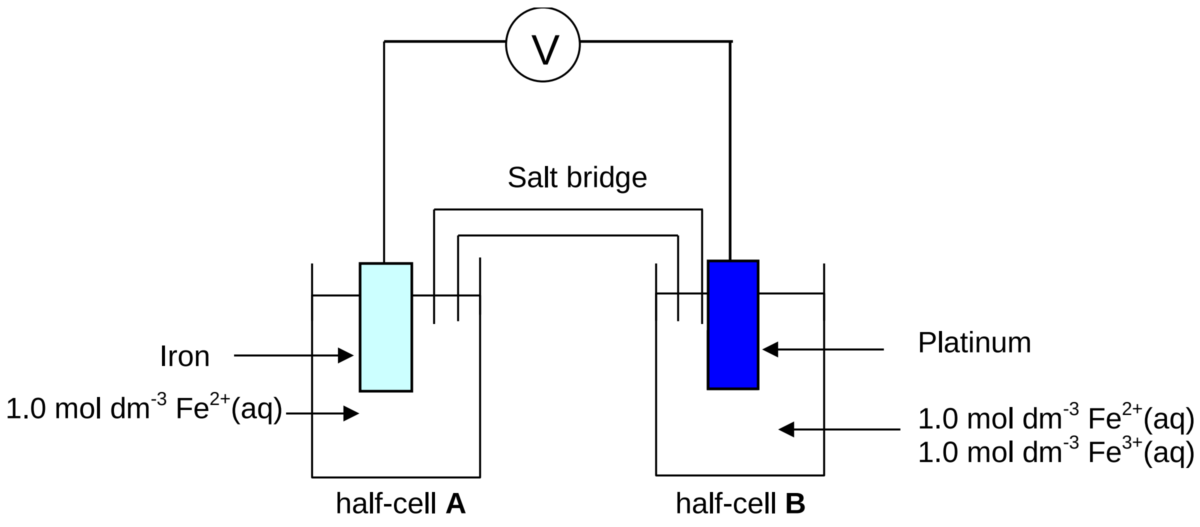
Moles of $H_2NCH(CH_3)COOH$ = 0.1y mol

Moles of H^+ remaining to react with 0.1y mol of $H_2NCH(CH_3)COOH$ = 0.2y – 0.1y = 0.1y mol

Hence, only the NH_2 group is protonated giving $H_3N^+CH(CH_3)COOH$

13 *Use of the Data Booklet is relevant to this question.*

The cell shown in the diagram below is set up, and the initial $E^{\ominus}_{cell}$ value was taken.



In experiment 1, some CN^- ions was added to half-cell **A**, before the $E^\ominus$ value was measured.
In experiment 2, excess CN^- ions was added to half-cell **B**, before the $E^\ominus$ value was measured.

How will the $E^\ominus_{cell}$ value in experiment 1 and 2 differ from the initial $E^\ominus_{cell}$ value?

	experiment 1	experiment 2
A	less positive	more positive
B	less positive	less positive
C	more positive	less positive
D	more positive	more positive

Answer:

$E_{cell} = E_{red} - E_{ox}$

In experiment 1,the equilibrium in the anode:

$$Fe^{2+} + 2e \rightleftharpoons Fe$$

will shift to the left when CN^- ions is added to the system as the $[Fe^{2+}]$ drops due to complex formation: $Fe^{2+} + 6CN^- \rightarrow [Fe(CN)_6]^{4-}$. E_{ox} becomes less positive and E_{cell} becomes more positive.

In experiment 2, the following equilibrium is set up in the cathode:

$$[Fe(CN)_6]^{3-} + e \rightleftharpoons [Fe(CN)_6]^{4-} \quad E_{red} = +0.36 \text{ V (from data booklet)}$$

The equilibrium has a E_{red} value lesser than before, $E_{red}(Fe^{3+}/Fe^{2+}) = +0.77V$. Hence E_{red} becomes less positive and E_{cell} becomes less positive.

The answer is **C**.

14 Which of the following elements has an oxide with a giant structure and a chloride which is readily hydrolysed?

A aluminium

B carbon

C phosphorus

D silicon

Answer:

Answer is **D**. SiO_2 exists as a giant covalent structure and $SiCl_4$ is hydrolysed in water to for SiO_2 and HCl.

15 For the elements in the third period of the Periodic Table, which property decreases consistently from sodium to chloride?

A electrical conductivity

B ionisation energy

C melting point

D radius of the atom

Answer:

Answer **A** is incorrect as the electrical conductivity increases from sodium to aluminum but drops after that.

Answer **B** is incorrect as the ionization energy increases steadily across the period due to increasing effective nuclear charge.

Answer **C** is incorrect as the melting point increases from sodium to aluminum but drops after that.

Answer **D** is correct as the effective nuclear charge increases across the period.

16 The carbonates of Group II decompose according to the following equation.

$$MCO_3(s) \longrightarrow MO(s) + CO_2(g)$$

For this reaction, ΔH increases on descending down the group.

carbonate	Mg	Ca	Sr	Ba
ΔH / kJ mol^{-1}	+101	+178	+235	+269

Which property best explains this trend?

A the ionic radius of the metal ion

B the ionisation energy of the metal

C the proton number of the metal

D the thermal stability of the oxide

Answer:

Answer is **A**. The thermal stability of the carbonates increases down the group for group II metals as the charge density of the cation decreases down the group making the extent of the polarisation of the C-O bonds in CO_3^{2-} get lesser and lesser. The charge density of the cation is in turn affected by the ionic radius of the metal ion.

17 Which of the following statements about the trends in the properties of the halogens down the group is correct?

A The volatility increases.

B The electronegativity increases.

C The reactivity as oxidising agents increases.

D The enthalpy change of reaction with hydrogen becomes more endothermic.

Solution:

Answer is D.

Option A is wrong. Moving down Group VII, electron cloud size increases and is more easily polarised. This leads to stronger van der waals (vdw) force of attraction between molecules, results in more energy required to break vdw, causing higher bolliling point. This means that volatility decreases.

Option B is wrong. The electronegativity decreases down Group VII as the valence electrons are further away from the nucleus due to the increase in number of principal quantum shell, results in increasing ease of losing valence electrons.

Option C is wrong. Moving down Group VII, the $E^{\ominus}$ of X_2/X^- becomes less positive, indicating that the oxidising power decreases.

Option D is correct. C-X bonds become longer and weaker in bond strength, the reaction becomes less exothermic (more endothermic).

18 In general, reactions of the compounds of transition elements can be classified under one or more of the following headings: acid-base, ligand exchange, precipitation, and redox.

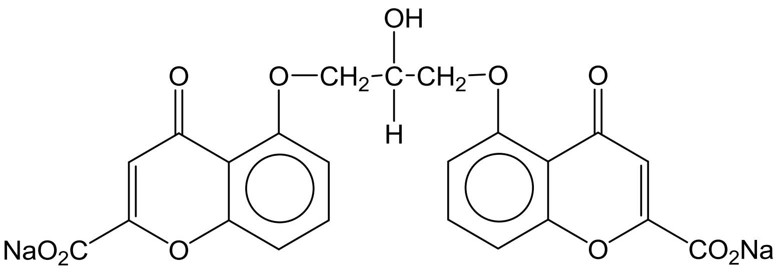
Which of the following classifications for the reactions is **incorrect**?

	reaction	acid-base	ligand exchange	precipitation	redox
A	$2FeCl_2 + Cl_2 \longrightarrow 2FeCl_3$				√
B	$[Cu(H_2O)_6]^{2+} + 4NH_3 \longrightarrow [Cu(NH_3)_4]^{2+} + 6H_2O$	√	√		
C	$[Cu(H_2O)_6]^{2+} + 4HCl \longrightarrow [CuCl_4]^{2-} + 4H^+ + 6H_2O$		√		
D	$[Fe(H_2O)_6]^{3+} + 3OH^- \longrightarrow Fe(OH)_3 + 6H_2O$	√		√	

Solution:
Answer is B.

Option B is wrong. It is just a ligand exchange reaction.

19 The anti-asthma drug *Intal* contains disodium cromoglycate which has the following structure.



How many chiral centres are there in the molecule?

A 0 **C** 1

B 2 **D** 3

Solution:
Answer is A.
There is no chiral carbon in this molecule and there is an internal plane of symmetry in the molecule.

20 The shape of unsaturated hydrocarbons can be explained in terms of hybridisation of orbitals.

Which bond is not present in 1-penten-3-yne, $CH_3C{\equiv}CCH{=}CH_2$?

A a π bond formed by 2p-2p overlap

B a σ bond formed by 1s-2sp overlap

C a σ bond formed by 2sp-2sp^2 overlap

D a σ bond formed by 2sp-2sp^3 overlap

Solution:

Answer is B.

Option A is correct. The π bond formed by 2p-2p overlap are those between C≡C and C=C.

Option C is correct. The bond formed between one of the carbons in C≡C and one of the carbons in C=C.

Option D is correct. The bond formed between the methyl group and one of the carbons in C≡C.

21 In the free radical substitution of 2-methylbutane with bromine, a mixture of mono-brominated compounds were obtained.

What is the statistical ratio of the two compounds with the highest yield?

A 1 : 1

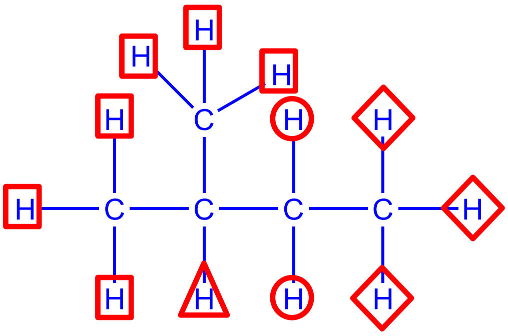
B 1 : 2

C 1 : 3

D 1 : 6

Solution:

Answer is B.



There are 4 different types of hydrogen in this molecule. (as shown above)

Therefore, the ratio of the two compounds with the highest yield is 3:6 = 1:2

22 In which of the following reactions is the inorganic reagent acting as a nucleophile?

A $C_6H_6 + HNO_3 \longrightarrow C_6H_5NO_2 + H_2O$

B $CH_3CH{=}CH_2 + Br_2 \longrightarrow CH_3CHBrCH_2Br$

C $CH_3CH_2NH_2 + HCl \longrightarrow CH_3CH_2NH_3Cl$

D $CH_3CH_2Br + NaOH \longrightarrow CH_3CH_2OH + NaBr$

Solution:

Answer is D.

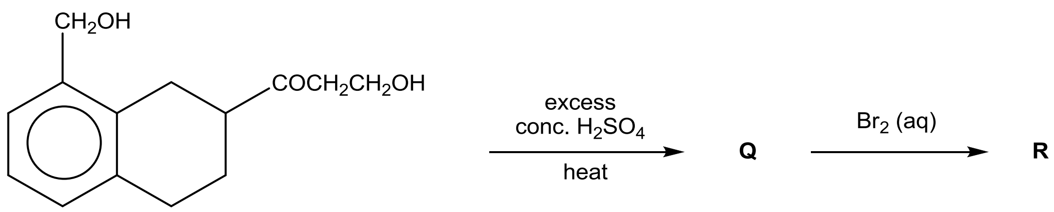
Option A, the nucleophile is the benzene (organic reagent).

Option B, the nucleophile is the alkene (organic reagent).

Option C, the nucleophile is the amine (organic reagent).

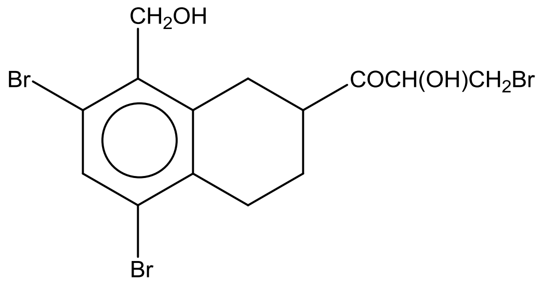
Option D, the nucleophile is the hydroxide ion (inorganic reagent).

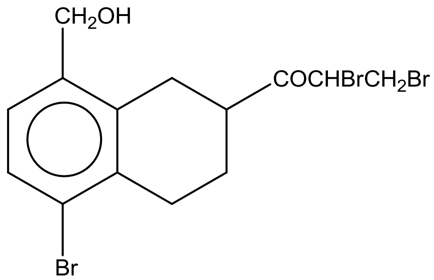
23 Compound **P** is used in the following synthesis.

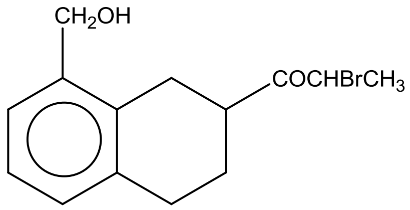


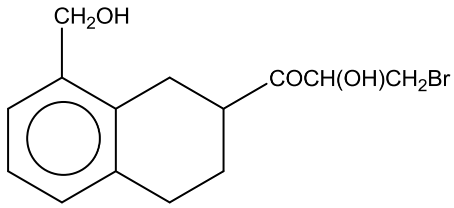
P

What is the likely structure of the final product **R**?

A 

B 

C 

D 

Solution:

Answer is D.

There are only 2 functional groups present in the molecule (ketone and alcohol).
The first reagent and condition is to undergo elimination to convert alcohol to alkene. The second reagent condition is the electrophilic addition of –OH and –Br across the C=C bond and option D is obtained. The benzyl alcohol remains intact throughout the reaction.

24 A bromine-containing organic compound, **T**, undergoes an elimination reaction when treated with hot ethanolic sodium hydroxide solution.

What is **T**?

A CH_3Br

B C_2Br_6

C $Br_2C{=}CH_2$

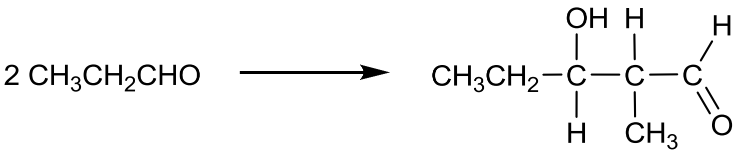
D $(CH_3)_2C{=}CBr_2$

Solution:

Answer is C.

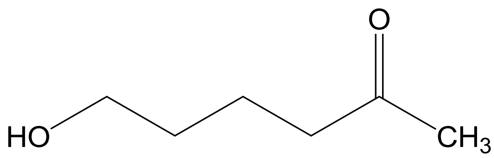
To undergo elimination, there must be a hydrogen atom on the carbon atom adjacent to the carbon atom bearing the halogen atom. Only option C fulfils this criterion.

25 The *aldol condensation* reaction occurs between carbonyl compounds. An example of the reaction is given below:

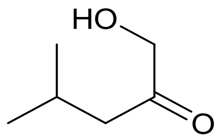


Which of the following compounds is the product of an aldol condensation reaction between 2 identical carbonyl compounds?

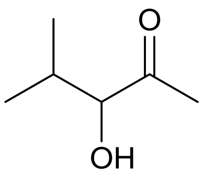
A



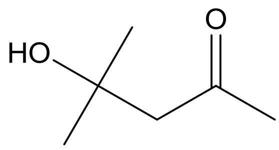
C



B

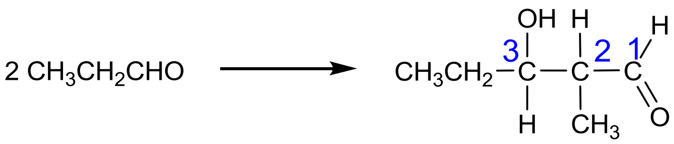


D

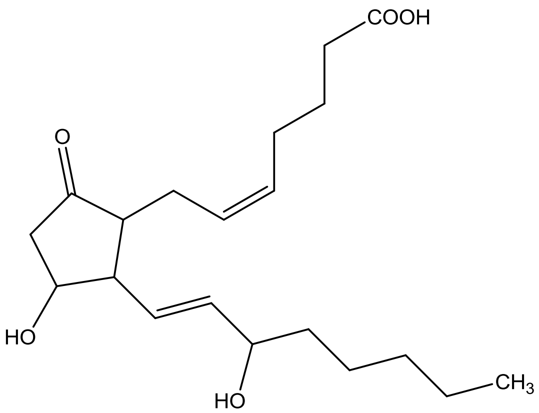


Ans: **D**

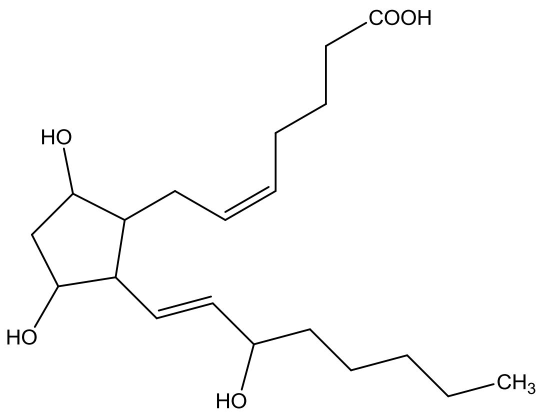
Through pattern recognition, the alcohol and carbonyl groups are at 1,3 positions to each other.



26 PGE_2 and $PGF_{2\alpha}$ are two prostaglandins, both with pharmacological activity.



PGE_2



$PGF_{2\alpha}$

Which reagent will convert PGE_2 into $PGF_{2\alpha}$ efficiently?

A H_2/Ni **C** $LiAlH_4$

B H_2O/H^+ **D** $NaBH_4$

Ans: **D**

From PGE_2 to $PGF_{2\alpha}$, only the ketone group is reduced to secondary alcohol. Thus, only $NaBH_4$ (specifically for reducing carbonyl groups) can be used.

H_2/Ni will reduce both ketone and alkene groups while $LiAlH_4$ will reduce both ketone and carboxylic acid groups. H_2O/H^+ is not a reducing agent; it is used for electrophilic addition of alkenes to form alcohols.

27 Which sequence shows the decreasing ease of hydrolysis?

A $C_6H_5Br > C_6H_5CH_2Cl > CH_3CH_2COCl > CH_3CH_2I$

B $CH_3CH_2COCl > CH_3CH_2I > C_6H_5CH_2Br > C_6H_5CH_2Cl$

C $C_6H_5Cl > CH_3COCH_2Cl > C_6H_5CH_2Br > CH_3CH_2I$

D $C_6H_5Br > CH_3CH_2Cl > C_6H_5CH_2I > CH_3CH_2COCl$

Ans: B

Relative ease of hydrolysis:

Acyl chloride ($RCOCl$) > halogenoalkane (RX) > halogenoarene (C_6H_5X)

Reason:

The C atom of the acyl chloride group is attached to 2 electron withdrawing groups, O and Cl as compared to the C atom of the halogenoalkane which is attached to only 1 electron withdrawing Cl group. Thus, C atom of the acyl chloride group is more electron deficient and thus more susceptible to nucleophilic attacks.

For halogenoarene, the p orbital of Cl overlaps with π orbitals of benzene ring, strengthening C–Cl, rendering it too strong to be broken.

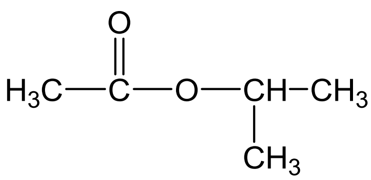
Within the halogenoalkane functional group,

Relative ease of hydrolysis:

RI < RBr < RCl

- Bond length of C–Cl < C–Br < C–I as atomic radius of I > Br > Cl
- Bond strength of C–Cl > C–Br > C–I
- Energy required to break C–X bond decreases in the order C–Cl > C–Br > C–I

28 Which one of the statements about the following ester is correct?



A It is a chain isomer of pentanoic acid.

B It is a functional group isomer of ethyl propanoate.

C It can be hydrolysed to produce an alcohol that can also be formed by the acid-catalysed hydration of propene.

D It can be hydrolysed to produce an alcohol that is resistant to oxidation by acidified potassium dichromate (VI).

Ans: **C**

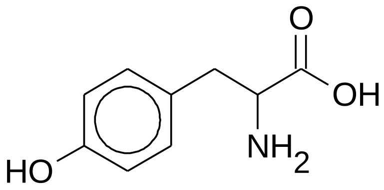
A: This statement is wrong as the ester is a functional group isomer of pentanoic acid.

B: This statement is wrong as the ester and ethyl propanoate have the same functional group.

C: The ester hydrolyses to give ethanoic acid and propan-2-ol. Propan-2-ol can also be formed from electrophilic addition of propene via steam/H^+ catalyst.

D: This is a wrong statement as the hydrolysed product, propan-2-ol can be oxidised by $K_2Cr_2O_7$ to form propanone.

29 Tyrosine is one of the 20 amino acids that is used by cells to synthesise proteins, and was first discovered in cheese.



Tyrosine

Which of the following types of reaction will Tyrosine **not** undergo?

A reduction

B nucleophilic substitution

C acid hydrolysis

D electrophilic substitution

Ans: **C**

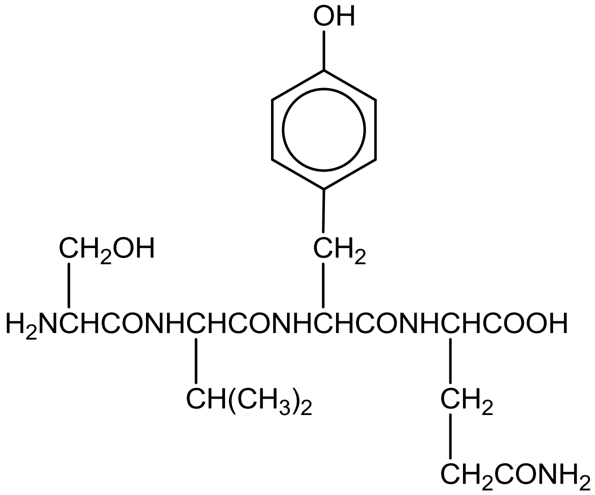
A: Carboxylic acid in tyrosine can undergo reduction to form primary alcohol.

B: Carboxylic acid and primary amine in tyrosine can undergo nucleophilic substitution with PCl_5(s) and halogenoalkane respectively.

C: There is no nitrile, amide or ester present in tyrosine to undergo acidic hydrolysis.

D: Phenol undergoes electrophilic substitution with X_2 or HNO_3.

30 Compound **X** formed from partial hydrolysis of the hormone insulin is shown below.



compound **X**

Which of the following statements is **incorrect**?

A Prolonged heating of compound **X** with dilute HCl produces 4 carbon-containing products.

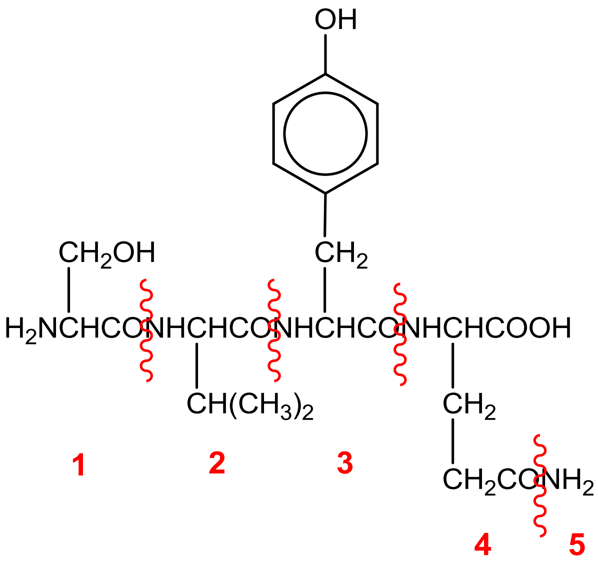
B Prolonged heating of compound **X** with dilute NaOH liberates an alkaline gas.

C 6 moles of NaOH is needed for complete reaction with 1 mole of compound **X**.

D 2 moles of Na_2CO_3 is needed for complete reaction with 1 mole of compound **X**.

Ans: **D**

Hydrolysing compound X breaks all the peptide linkages.



A: See diagram above for the 4 carbon-containing products.

B: Alkaline hydrolysis produces NH_3(g) – fragment 5.

C: 5 moles of NaOH(aq) is required for hydrolysing the peptide bonds and 1 mole of NaOH is for acid base with phenol group.

D: Na_2CO_3 reacts with carboxylic acid group only. Thus, only 1 mole of Na_2CO_3 is needed.

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

31 Two ions formed by the isotopes of iron have chemical formulae shown below:

$$^{54}_{26}Fe^{2+} \qquad ^{56}_{26}Fe^{2+}$$

Which of the following statements about these two ions are **incorrect**?

1 The electronic configuration of both Fe^{2+} is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$.

2 $^{56}_{26}Fe^{2+}$ gives a greater angle of deflection than $^{54}_{26}Fe^{2+}$ in an electric field.

3 The total number of protons and electrons in each of the Fe^{2+} ions is smaller than the nucleon number of the respective ion.

Ans: B (1 and 2 are correct)

1: The electronic configuration of both Fe^{2+} ions is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$.
2: Angle of deflection is directly proportional to charge/ mass. Since $^{56}_{26}Fe^{2+}$ and $^{54}_{26}Fe^{2+}$ have the same charge, but the former has a larger mass, thus $^{56}_{26}Fe^{2+}$ has a lower angle of deflection.
3: Total number of protons and electrons in $^{54}_{26}Fe^{2+}$ = 26 + 24 = 50 which is lesser than the nucleon number 54.
Total number of protons and electrons in $^{56}_{26}Fe^{2+}$ = 26 + 24 = 50 which is lesser than the nucleon number 56.

32 The definitions of many chemical terms can be illustrated by chemical equations.

Which terms can be illustrated by an equation that shows the formation of a positive ion?

1 first ionisation energy

2 heterolytic fission

3 electron affinity

Ans: B (1 and 2 are correct)

1. First ionisation energy: X(g) → X^+(g) + e

2. Heterolytic fission produces a cation and anion e.g. *Cl*—*Cl* → Cl^+ + Cl^-

3. Electron affinity: X(g) + e → X^-(g)

33 Which of the following mixtures would constitute a buffer solution?

1 1 mol of $H_2PO_4^-$ and 1 mol of HPO_4^{2-}

2 1 mol of HC*l* and 2 mol of NH_3

3 1 mol of NaOH and 2 mol of 1,4-benzenedicarboxylic acid

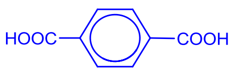
Ans: **A** (**1**, **2** and **3** are correct).

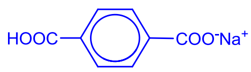
Recall: A buffer is comprised of:
- a weak acid & its conjugate base OR
- a weak base & its conjugate acid.

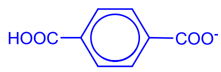
Option **1:** Weak acid is $H_2PO_4^-$, its conjugate base is HPO_4^{2-}. Hence, it is a buffer.

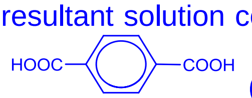
Option **2**: $HCl + NH_3 \rightarrow NH_4^+Cl^-$

Based on mole ratio, the resultant solution contains 1 mol of NH_4^+ (product of the reaction) and 1 mol of NH_3 (unreacted, since NH_3 is in excess).
Weak base is NH_3, its conjugate acid is NH_4^+. Hence, it is a buffer.

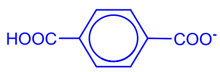
Option **3:** NaOH + 

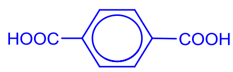
→ H_2O + 

Based on mole ratio, the resultant solution contains 1 mol of 

(product of the reaction) and 1 mol of 

(unreacted, since it is in excess).

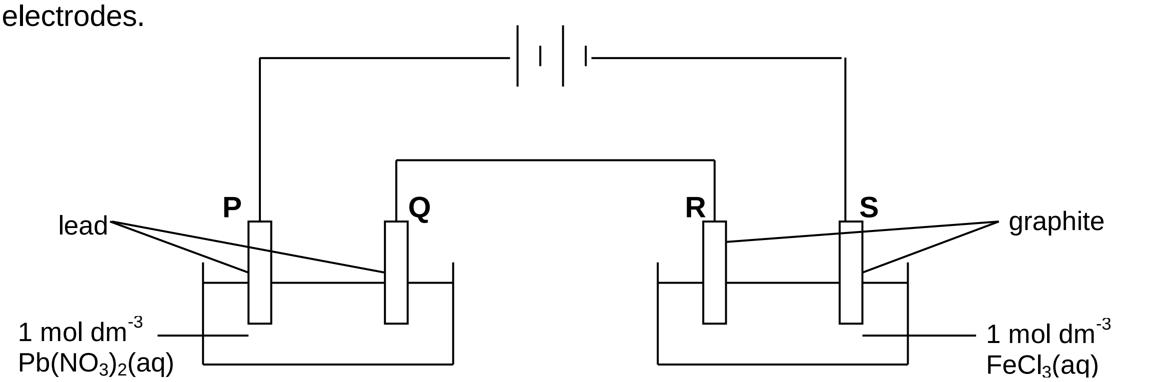
Weak base is 

, its conjugate acid is 

. Hence, it is a buffer.

34 *Use of the Data Booklet is relevant to this question.*

Two electrolytic cells are connected in series as shown in the diagram where **P**, **Q**, **R** and **S** are electrodes.



Which statements about the products formed at each electrode are correct?

1 Fe^{2+} is formed at electrode **S**.

2 H_2 is formed at electrode **Q**.

3 O_2 is formed at electrode **P**.

Ans: **D** (**1** only is correct).

Note: The electrodes for the cell on the left are reactive (may be oxidised/reduced).

$Fe^{3+} + e^-$	$\rightleftharpoons$	Fe^{2+}	+0.77
$2H_2O + 2e^-$	$\rightleftharpoons$	$H_2 + 2OH^-$	–0.83

For electrode **S**, which is cathode (-ve), Fe^{3+} will reduced to Fe^{2+} (since it has more +ve $E_{Fe3+|Fe2+}$ than $E_{H2O|H2}$). Hence, option **1** is correct.

$Pb^{2+} + 2e^-$	$\rightleftharpoons$	Pb	–0.13
$2H_2O + 2e^-$	$\rightleftharpoons$	$H_2 + 2OH^-$	–0.83

For electrode **Q**, which is cathode (-ve), Pb^{2+} will be reduced to Pb (since it has more +ve/ less -ve $E_{Pb2+|Pb}$ than $E_{H2O|H2}$). Hence, option 2 is wrong.

$Pb^{2+} + 2e^-$	$\rightleftharpoons$	Pb	–0.13
$O_2 + 4H^+ + 4e^-$	$\rightleftharpoons$	$2H_2O$	+1.23

For electrode **P**, which is anode (+ve), Pb will be oxidised to Pb^{2+} (since it has more –ve $E_{Pb2+|Pb}$ than $E_{O2|H2O}$). Hence, option 3 is wrong.

35 The Group II metals have higher melting points than the Group I metals.

Which factors could contribute towards the higher melting points?

1 There are smaller interatomic distances in the metallic lattices of the Group II metals.

2 Two valence electrons are available from each Group II metal atom for bonding the atom into the metallic lattice.

3 Group II metals have the higher first ionisation energy.

Ans: **B** (**1** and **2** are correct).

Recall:
- Energy is needed to overcome the metallic bonds during melting of metals
- Metallic bond strength is dependent on:
  (1) charge density,
  (2) number of valence e^- contributed to delocalised e^-

Option **1** is correct – As the interatomic distances is smaller in Group II metals, the charge density (since radius is smaller) is higher, hence, stronger metallic bond and higher mp.

Option **2** is correct – As Group II metals contribute 2 valence e to the sea of delocalised e-, it has stronger metallic bond, and hence, higher mp.

Option **3** is incorrect – Higher first IE relates to metals in gaseous state. Hence, does not explain the mp.

36 *Use of the Data Booklet is relevant to this question.*

Which pairs of reagents when reacted together would produce a change in the oxidation numbers of the metal atoms?

1 aqueous potassium manganate(VII) and acidified aqueous iron(II) sulfate

2 aqueous acidified lead(IV) chloride and aqueous vanadium(III) nitrate

3 aqueous ammonia and an aqueous suspension of silver chloride

Ans: **B** (**1** and **2** are correct).

Note: Change in oxidation numbers ➔ redox reaction ➔ E values from Data Booklet

Option 1: (Note: $KMnO_4$ is an oxidising agent➔ Fe^{2+} may be oxidised to Fe^{3+})

$$Fe^{3+} + e^- \rightleftharpoons Fe^{2+} \qquad +0.77$$
$$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O \qquad +1.52$$

$MnO_4^- + 5Fe^{3+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{2+} + 4H_2O$

$E^{\ominus}_{cell}$ = (+1.52) – (+0.77) = +0.75 > 0 (reaction is energetically feasible)

Oxidation number of Mn changes from +7 in MnO_4^- to +2 in Mn^{2+}, while oxidation number of Fe changes from +3 in Fe^{3+} to +2 in Fe^{2+}. Option **1** is correct.

Option **2**: (Note: V^{3+} may only be oxidised, hence, Pb_4^+ may be reduced; Cl^- cannot be reduced any further)

$$Pb^{4+} + 2e^- \rightleftharpoons Pb^{2+} \qquad +1.69$$
$$VO^{2+} + 2H^+ + e^- \rightleftharpoons V^{3+} + H_2O \qquad +0.34$$

$Pb^{4+} + 2V^{3+} + 2H_2O \rightarrow Pb^{2+} + 2VO^{2+} + 4H^+$

$E^{\ominus}_{cell}$ = 1.69 – 0.34 = +1.35V > 0 (reaction is energetically feasible)

** VO^{2+} may be further oxidised to VO_2^+ in acidic medium.

$Pb^{4+} + 2VO^{2+} + 2H_2O \rightarrow Pb^{2+} + 2VO_2^+ + 4H^+$

$E^{\ominus}_{cell}$ = 1.69 – 1.00 = +0.69V (> 0, reaction is energetically feasible)

$PbCl_4$ will oxidise V^{3+} to VO^{2+} and then to VO_2^+ in an acidic medium.

Oxidation number of Pb changes from +4 in Pb_4^{+-} to +2 in Pb^{2+}, while oxidation number of V changes from +3 in V^{3+} to +4 in VO^{2+}. Option **2** is correct.

Option **3**: AgCl is soluble in NH_3 to form diamminesilver(I) ion (Recall: Group VII)

$Ag^+(aq) + Cl^-(aq) \rightleftharpoons AgCl(s)$

$Ag^+(aq) + 2NH_3(aq) \rightleftharpoons [Ag(NH_3)_2]^+$

This is a ligand exchange reaction. There is no change in oxidation number for Ag.

37 Which reactions yield a carbon compound incorporating deuterium, D? [D = ^{2}H]

1 $CH_3COCI_3 \xrightarrow[D_2O]{NaOD}$

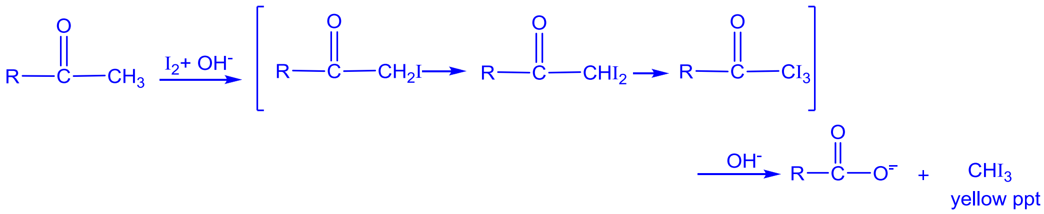
2 $CH_3CH_2CN \xrightarrow[D_2O]{NaOD}$

3 $(CH_3)_3COH \xrightarrow[\text{heat}]{\text{conc. } D_2SO_4}$

Ans: **D** (**1** only is correct).

Option **1**: This is mild oxidation reaction. The product contains deuterated tri-iodomethane (CDI_3).

Recall (from your lecture notes):



At the **<u>last step</u>** of the reaction (using OD^- instead of OH^-),

$CH_3COCI_3 \xrightarrow{\text{mild oxidation}} CH_3COO^- + CDI_3$

The carbon containing compounds contains D atoms. **1** is correct.

Option **2**: This is alkaline hydrolysis of nitrile compounds.

$CH_3CH_2CN \xrightarrow{\text{Alkaline hydrolysis}} CH_3CH_2COO^-$

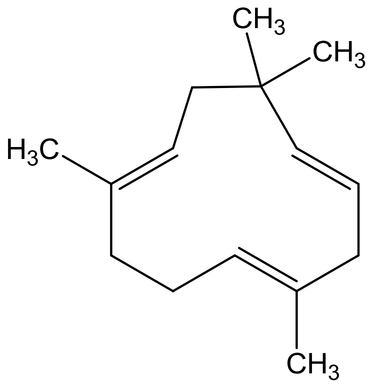
The carbon containing compound does not contain D atoms. **2** is incorrect.

Option 3: This is elimination of alcohol.

$(CH_3)_3COH \xrightarrow{\text{Elimination}} (CH_3)_2C{=}CH_2$

The carbon containing compound does not contain D atoms. **3** is incorrect.

38 Humulene can be extracted from carnation flowers.



humulene

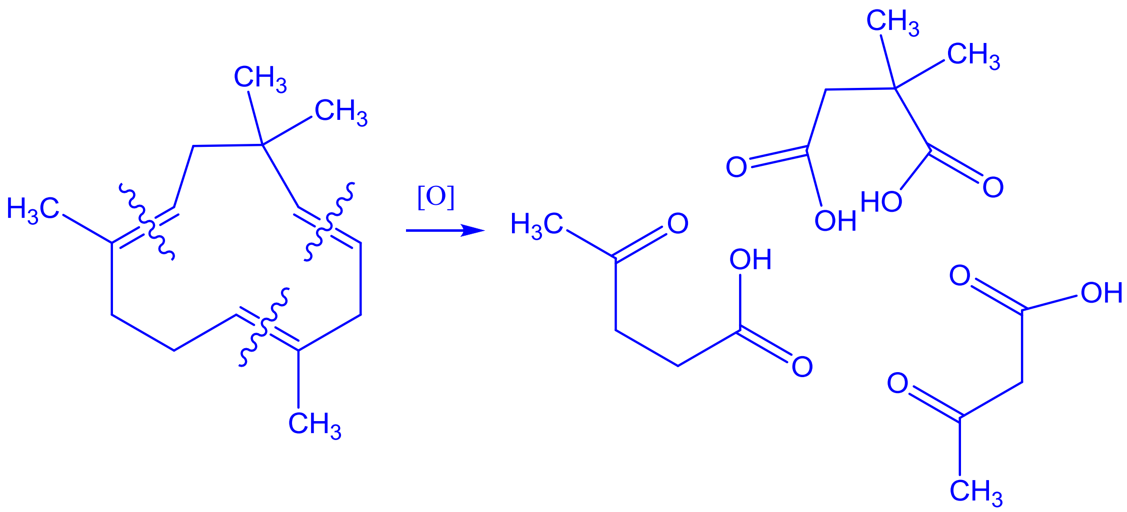
Which products are obtained from the reaction of humulene with hot acidified concentrated $KMnO_4$?

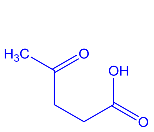
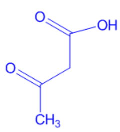
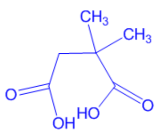
1 $CH_3COCH_2CH_2CO_2H$

2 $CH_3COCH_2CO_2H$

3 $HO_2CCH_2C(CH_3)_2CO_2H$

Ans: **A** (**1**, **2** and **3** are correct).

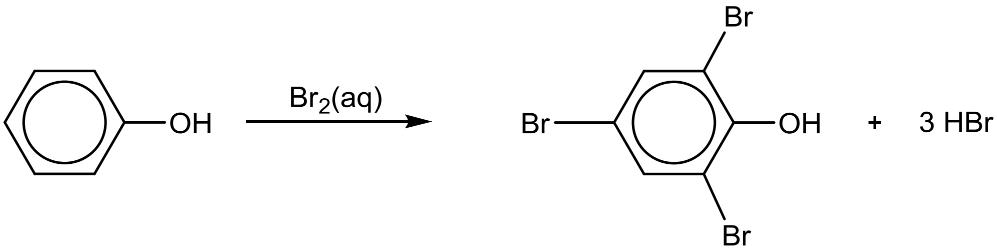


1 is , **2** is  and **3** is . Hence, all are correct.

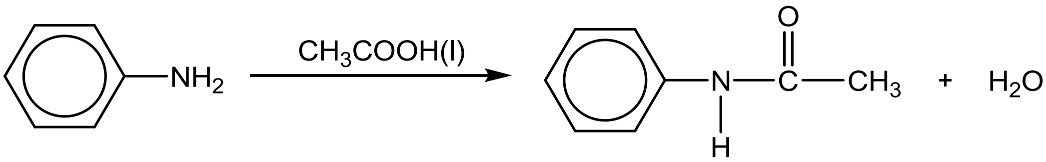
39 Three different transformations are given.

Which transformations are correct?

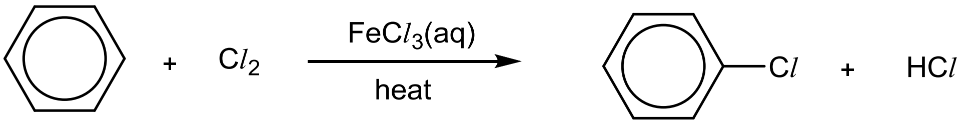
1



2



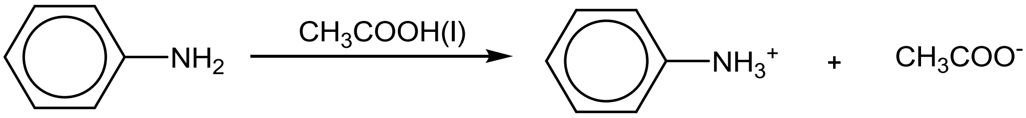
3



Ans: **D** (**1** only is correct)

Option **1** is correct.

Option **2** is wrong. Phenylamine undergoes acid-base reaction with carboxylic acid instead.



To form the amide, acyl chloride should be used instead.

Option **3** is wrong. The reaction requires anhydrous $FeCl_3$(s).

40 Keratin, a family of fibrous protein, is a key structural component in hair. The curliness of the hair is determined by the number of disulfide bonds that can be formed between R groups ($–CH_2SH$) of the cysteine residues of adjacent keratin molecules. The formation of disulfide bonds can be represented by the following equation:

$$2\ –CH_2SH \longrightarrow –CH_2S–SCH_2– + 2H^+ + 2e$$

With more disulfide bonds present, the hair is curlier.

In hair rebonding, heat or chemical treatment may be applied to curly hair to straighten the hair.

Which of the following are true?

1 For hair rebonding using heat treatment, the disulfide bonds are overcomed.

2 An oxidising agent may be used in the chemical treatment to straighten the hair.

3 The primary structure of keratin is disrupted during hair rebonding.

Ans: **D** (**1** only is correct)

Option **1** is true. Heating breaks the disulfide bonds, resulting in a straighter hair.

Option **2** is false. The formation of disulfide bonds is a oxidation (loss of electron). In straightening hair (breaking of disulfide bonds), reduction occurs. Hence, a reducing agent is required.

Option **3** is false. Heating or chemical treatment to straighten hair overcomes the R groups interactions. Tertiary structure is disrupted, but primary structure is not.

Final Answer

1	B	**6**	C	**11**	A	**16**	A	**21**	B	**26**	D	**31**	B	**36**	B
2	C	**7**	B	**12**	D	**17**	D	**22**	D	**27**	B	**32**	B	**37**	D
3	D	**8**	A	**13**	C	**18**	B	**23**	D	**28**	C	**33**	A	**38**	A
4	B	**9**	A	**14**	D	**19**	A	**24**	C	**29**	C	**34**	D	**39**	D
5	B	**10**	A	**15**	D	**20**	B	**25**	D	**30**	D	**35**	B	**40**	D